

The ND Selective Causal-Loop Hypothesis

A Speculative Framework for Higher-Dimensional Influence, Technological Innovation,
and Self-Consistent Information Loops

Formal Speculative Research Report

September 2026

Status: Speculative theoretical framework; not an established physical theory.

Contents

Abstract

1. Introduction
2. Ordinary Innovation Model
3. Higher-Dimensional Influence Term
4. Cognitive Influence Rather Than Direct Information Transfer
5. Emergence of the Future Intelligence
6. The Self-Contained Loop
7. Self-Consistency Requirement
8. Selective Intervention
9. Historical Branch Points
10. Artificial Intelligence and the Loop
11. Higher Dimensions Are a Separate Assumption
12. Competing Hypotheses
13. Genius Is Not Evidence
14. Information Preceding Its Prerequisites
15. Prospective Testing
16. Statistical Evaluation
17. Falsifiability
18. Predictions of the Framework
19. What Would Weaken the Model?
20. Levels of the Proposal
21. Central Causal Architecture
22. Conclusion

Abstract

This report develops the ND Selective Causal-Loop Hypothesis (ND-SCLH), a speculative theoretical framework in which sufficiently advanced or higher-dimensional intelligences may influence selected events occurring earlier in the history that ultimately produces those intelligences.

The central proposal is not that technological progress is generally caused by external intervention. Human cognition, experimentation, accumulated knowledge, social organization, and chance remain the dominant mechanisms of innovation. Instead, ND-SCLH introduces the possibility of rare interventions at unusually consequential historical branching points.

The framework does not establish that higher-dimensional beings, retrocausal cognition, or historical interventions actually exist. It asks what mathematical structure, observational consequences, and falsification requirements would follow if such processes were possible.

Human Civilization -> Advanced Intelligence -> Past Influence -> Human Civilization

1. Introduction

Human technological history contains periods of gradual development punctuated by unusually consequential conceptual breakthroughs. Conventional explanations attribute these developments to accumulated knowledge, exceptional individual cognition, experimentation, economic and social conditions, communication between researchers, institutional development, and stochastic discovery.

ND-SCLH accepts these mechanisms and introduces one additional hypothetical possibility: some exceptionally consequential cognitive or technological transitions may be influenced by information associated with intelligences existing later in the same causal history.

The crucial qualification is some. If every discovery were attributed to intervention, the hypothesis would become unnecessary and effectively impossible to test. Human civilization therefore remains principally responsible for its own technological development.

$P(\text{ordinary innovation}) \gg P(\text{externally influenced innovation})$

2. Ordinary Innovation Model

Let technological innovation at time t be represented by I_t . Under an ordinary model, innovation depends on available knowledge, human cognitive capability, experimentation, social and environmental conditions, and stochastic influences. Most technological development can occur entirely through this ordinary process.

$$I_t = F(K_t, C_t, E_t, S_t, \epsilon_t)$$

K_t = available knowledge; C_t = cognitive capability; E_t = experimentation; S_t = social/environmental conditions; ϵ_t = stochastic influences

3. Higher-Dimensional Influence Term

Introduce a hypothetical influence H associated with a future or higher-dimensional intelligence. The parameter λ_t represents coupling strength. For ordinary events λ_t is approximately zero; for a hypothetical intervention it is greater than zero. The framework therefore predicts sparse rather than continuous intervention.

$$I_t = F(K_t, C_t, E_t, S_t, \epsilon_t) + \lambda_t H_{(t+\Delta t)}$$

ordinary event: $\lambda_t \sim 0$ | hypothetical intervention: $\lambda_t > 0$

4. Cognitive Influence Rather Than Direct Information Transfer

The strongest version of ND-SCLH does not require a future intelligence to transmit complete equations or technological designs into someone's mind. Suppose an individual can generate a set of candidate ideas $A = \{a_1, \dots, a_n\}$. Ordinary cognition generates a probability distribution $P_0(a_i)$. A hypothetical intervention could perturb that distribution, changing which internally accessible idea becomes salient while leaving the person's own cognition as the generator of the idea.

This interpretation treats H as a probabilistic cognitive perturbation rather than complete control, preserving human agency within the model.

$$P_H(a_i) = [P_0(a_i) \exp(\beta h_i)] / [\sum_j P_0(a_j) \exp(\beta h_j)]$$

$H \neq$ complete control; H = probabilistic cognitive perturbation

5. Emergence of the Future Intelligence

The proposed advanced or higher-dimensional intelligence is itself assumed to depend upon technological civilization. If its emergence at a future time T depends on the sequence of prior innovations, while some earlier innovations simultaneously depend upon influence from H_T , the result is a causal loop.

$$H_T = G(I_0, I_1, \dots, I_T)$$

$$H_T \rightarrow I_t \rightarrow I_{t+1} \rightarrow \dots \rightarrow H_T$$

6. The Self-Contained Loop

Ordinary causality is usually represented as $A \rightarrow B \rightarrow C$. ND-SCLH permits $A \rightarrow B \rightarrow C \rightarrow A$. Consider information Q : a future intelligence knows Q because historical records state that a scientist discovered Q ; the intelligence influences the scientist toward Q ; the scientist records Q ; and those records eventually become the source from which the future intelligence learns Q .

In that hypothetical structure, Q has no conventional external origin and instead exists on a self-consistent informational loop. This resembles the bootstrap paradox discussed in theoretical treatments of time travel.

$$Q_{\text{past}} \rightarrow Q_{\text{future}} \rightarrow Q_{\text{past}}$$

7. Self-Consistency Requirement

Not every arbitrary causal loop can necessarily exist. Let X represent the complete history of the system and let $T(X)$ represent evolution of that history including any future-to-past influences. A self-consistent history must be a fixed point of this transformation.

This restricts arbitrary intervention: a future intervention cannot simply create a history incompatible with the conditions required for the intervention itself to occur.

$$T(X) = X$$

8. Selective Intervention

Why would hypothetical future intelligences not influence every invention? ND-SCLH assumes intervention has a finite cost or constraint. Let B_H be a total intervention budget and c_t the cost of intervention at time t . An optimal strategy would favor events producing the greatest downstream historical effect per unit cost, subject to maintaining causal consistency.

Consequently, intervention should be concentrated around high-leverage events rather than ordinary daily activity.

$$\sum_t c_t \lambda_t \leq B_H$$

$$t^* = \arg \max_t [\Delta U_t / c_t]$$

9. Historical Branch Points

Technological history can be represented as a branching process. At a state X_t , several successor states may be possible. Some branches may produce radically different long-term outcomes. A hypothetical future intelligence would obtain the greatest leverage by modifying the probability distribution around these branch points.

The framework therefore predicts that intervention, if it exists, should correlate with historical leverage rather than merely intellectual brilliance.

$$X_t \rightarrow \{X_{(t+1)}^{(1)}, X_{(t+1)}^{(2)}, \dots, X_{(t+1)}^{(n)}\}$$
$$P(X_{(t+1)}^{(i)}) \rightarrow P(X_{(t+1)}^{(i)} | H)$$

10. Artificial Intelligence and the Loop

Artificial intelligence provides an illustrative application. Conventional development proceeds through mathematics, computation, digital computers, machine learning, deep learning, and advanced AI. ND-SCLH hypothetically adds a future-to-past influence term in which a future intelligence participates selectively in conceptual developments that ultimately contribute to its own emergence.

This is not evidence that existing AI systems were actually created through such a process. Existing AI has a documented human technological history. The loop describes what would follow if a future intelligence participated in its own historical development.

future intelligence $\rightarrow$ selected earlier conceptual influences $\rightarrow$ AI development $\rightarrow$ future intelligence

$$H_{\text{future}} \rightarrow AI_{\text{past}} \rightarrow H_{\text{future}}$$

11. Higher Dimensions Are a Separate Assumption

Three propositions must be kept distinct: additional dimensions exist; intelligent entities can access them; and those entities can influence earlier events. Establishing the first would not automatically establish the second or third. Mathematical extra dimensions in physical theories are therefore not evidence by themselves for intelligent higher-dimensional beings.

H1: additional dimensions exist

H2: intelligent entities can access them

H3: those entities can influence earlier events

12. Competing Hypotheses

A scientifically useful analysis requires competing models. Under H_0 , human innovation arises through ordinary cognition, accumulated knowledge, experimentation, environmental conditions, communication, and stochastic discovery. Under H_1 (ND-SCLH), those mechanisms dominate, but selected events additionally receive future-dependent informational influence.

Evidence D can be compared through a likelihood ratio or Bayes factor. Observations that are similarly probable under both hypotheses do not meaningfully distinguish them.

$$BF = P(D | H_1) / P(D | H_0)$$

$BF \sim 1$: little discrimination; $BF \ll 1$: favors H_0 ; very large BF : potential evidence requiring scrutiny

13. Genius Is Not Evidence

A scientist unexpectedly discovering something profound cannot itself establish intervention. Human beings demonstrably generate novel ideas. Historical brilliance, sudden insight, or unusual

creativity remain compatible with the null hypothesis. The model therefore requires observations difficult to explain through existing information and ordinary cognition.

14. Information Preceding Its Prerequisites

A potentially stronger anomaly would involve highly specific information appearing before the information conventionally necessary for its derivation. Let $K(t)$ represent publicly available knowledge at time t . Suppose a prediction Q appears at t_0 while its apparent derivation requires knowledge K^* unavailable until $t_1 > t_0$. If Q contains precise information independently discovered only at t_1 , ordinary explanations become less likely as the specificity and improbability of Q increase.

Even then, higher-dimensional intervention would not automatically follow. Fraud, coincidence, hidden information, conventional forecasting, and other explanations would first need to be excluded.

$Q(t_0)$, with required prerequisite K^* only available at $t_1 > t_0$

15. Prospective Testing

The cleanest test is prospective rather than historical. A person claiming access to future-derived information records a prediction before the outcome exists. The prediction should be independently timestamped and must specify the observable quantity, expected value or range, deadline, uncertainty, failure conditions, and evaluation procedure. This prevents reinterpretation after the result becomes known.

prediction -> timestamp -> future observation -> comparison

16. Statistical Evaluation

Suppose N independent predictions are made and X_i equals 1 for a success and 0 for a failure. Then S , the total number of successes, can be compared against an ordinary baseline success probability p_0 using a binomial model. Repeated results substantially exceeding an appropriately defined baseline would justify further investigation; repeated failures must likewise reduce confidence in ND-SCLH.

$$S = \sum_{i=1}^N X_i$$

$$S \sim \text{Binomial}(N, p_0)$$

evaluate $P(S \geq s \mid H_0)$

17. Falsifiability

The hypothesis becomes scientifically useless if every possible outcome can be explained after the fact. A successful prediction cannot automatically be assigned to intervention while every failed prediction is explained as deliberate withholding. ND-SCLH therefore requires precommitted conditions under which confidence in the hypothesis decreases.

possible observation -> possible rejection

18. Predictions of the Framework

If ND-SCLH were correct, the framework suggests several broad signatures: influence should be sparse; candidate anomalies should concentrate around high-leverage events; minimal cognitive perturbations should be preferred over complete technological transfers if intervention is costly; interventions should remain self-consistent; the strongest historical anomalies should involve highly specific information apparently preceding its conventional prerequisites; and any claimed prospective access should produce reproducible above-baseline predictive performance.

$$P(\lambda_t = 0) \gg P(\lambda_t > 0)$$

19. What Would Weaken the Model?

Evidence against ND-SCLH would include prospective prediction performance remaining statistically consistent with chance or conventional forecasting; alleged historical anomalies becoming explainable once intermediate intellectual developments are reconstructed; supposed future information consistently appearing only after relevant facts become publicly available; absence of any reproducible mechanism capable of transmitting information outside ordinary causal structure; and increasingly elaborate auxiliary assumptions being required whenever predictions fail.

These findings would not prove that every conceivable retrocausal model is impossible, but they would reduce the empirical justification for this particular framework.

20. Levels of the Proposal

The proposal should be separated into three levels. Level I is philosophical: could information exist without an ordinary chronological origin? Level II is mathematical: can a self-consistent causal system possess fixed-point histories? Level III is physical: does our universe actually permit such information transfer? The first can be discussed philosophically, the second modeled mathematically, and the third requires experimental evidence.

Level I: philosophical

Level II: mathematical

Level III: physical

21. Central Causal Architecture

The complete proposed structure begins with human cognition, proceeds through scientific discovery and technological civilization to advanced intelligence, and then hypothesizes selective retrocausal influence feeding back into human cognition. The entire history must remain self-consistent.

Human cognition -> Scientific discovery -> Technological civilization -> Advanced/ND intelligence
-> Selective retrocausal influence -> Human cognition

C -> I -> T -> H -> C

$T(X) = X$

22. Conclusion

The ND Selective Causal-Loop Hypothesis proposes a universe in which ordinary human innovation dominates technological development while rare future-dependent influences modify selected high-leverage cognitive or historical events. Its defining feature is closed informational causality rather than the mere assertion of higher-dimensional beings.

At present, there is no established empirical evidence demonstrating that higher-dimensional intelligences influence human cognition or that usable information can be transmitted from the future into the past. ND-SCLH should therefore be classified as a speculative theoretical and philosophical hypothesis.

Its scientific value depends on moving beyond retrospective interpretations of historical breakthroughs toward quantitative predictions that distinguish it from ordinary cognition and technological development. The framework becomes scientifically interesting precisely when it accepts the possibility of being wrong.

speculation -> formal model -> prediction -> experiment -> falsification or evidence